

Masking Strategies for nano4M



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Project page: <https://rhabib22.github.io/nano4M-masking-strategies/>

Abstract—We study how the choice of masking strategy at pre-training affects what a small multimodal model learns. Building on nano4M, we train five variants on CLEVR with identical architecture, data, compute and seeds, varying only the masking strategy: random (V0), 2D block (V1), context-block (V2), text-side contiguous-span (V3), and a per-sample mix of V0 and V2 (V4). We evaluate each variant under a unified protocol across reconstruction and cross-modal generation, scoring with a parser, CIDEr, pixel-level similarity, an LLM judge and an open-vocabulary detector. V2 dominates the meaning-aware and object-grounded metrics in both generation directions. V1 wins pixel-level reconstruction. V4 wins lexical caption similarity in image-to-caption, a metric neither of its constituent strategies leads alone. V3 ties with the random baseline on every text metric, which we report as an honest negative result attributable to the visual-grounding bottleneck in our four-modality setup.

I. INTRODUCTION

Multimodal masked models like 4M [1] learn to fill in hidden tokens across multiple modalities, and the masking strategy at pre-training is the only thing that defines this self-supervised objective. Random masking is the de facto choice, but it has a subtle weakness: when a token is hidden, its immediate neighbours in the same modality usually stay visible. The model can solve much of its training loss by predicting from local-modality context alone, which means cross-modal binding (using depth to predict text, or text to predict pixels) stays optional rather than necessary.

This raises an empirical question: if random masking lets the model skip cross-modal binding, does a more constrained mask force the model to learn fundamentally different skills? And do those skills transfer to downstream cross-modal generation tasks?

We test this on nano4M, a small four-modality version of 4M trained on CLEVR [2]. We train five variants under matched conditions: identical FourM backbone (encoder and decoder depth 6, model dimension 512), identical 5-billion-token budget, identical optimizer, identical seeds; varying only the masking strategy. V0 is the random baseline. V1 masks contiguous square blocks of image tokens, removing the local-neighbour shortcut. V2 keeps a single small visible block per image modality, hiding around 90% of every

image-side modality and leaving the caption as the only substantially-visible scaffold. V3 masks contiguous spans of the caption itself, calibrated to CLEVR field length. V4 is a per-sample 50/50 mixture of V0 and V2. Six single-knob ablations test the sensitivity of V1 and V2 to their block-size hyperparameter.

We evaluate every variant under a unified protocol on a held-out CLEVR val subset (3 seeds, 500 samples per seed), covering per-modality reconstruction and cross-modal generation in both directions. Our headline finding is that different masks produce specialists: V2 wins meaning-aware and object-grounded metrics, V1 wins pixel-level reconstruction, V4 wins the lexical caption-similarity metric in image-to-caption, and V3 ties with V0. The mask, in our setting, is the objective.

II. RELATED WORK

Random multimodal masking. The 4M family [1] uses a unified random masking objective across modalities, with a per-sample Dirichlet allocation of tokens between encoder input and decoder target. This is the baseline we compare against (our V0). The 4M line of work focuses on scaling and on extending the modality set; we hold architecture and data fixed and treat the masking strategy itself as the independent variable.

Block masking for vision. Block masking, where contiguous square regions of image tokens are hidden rather than scattered positions, has been studied in masked-autoencoder work [3] and in generative image transformers [4]. MaskGIT [4] also introduced parallel iterative decoding for masked image generation; we use the same eight-step parallel decoding scheme at evaluation for all variants. Our V1 isolates the effect of block masking as a training-side objective in a multimodal setting.

Span corruption for text. T5 [5] introduced contiguous-span masking as a text pre-training objective for language models, and demonstrated transfer to downstream tasks when followed by task-specific fine-tuning. Our V3 applies the same span structure to the text modality inside a multimodal masked model, using standard parallel decoding to keep the comparison apples-to-apples with the other variants.

Our contribution. We provide a controlled five-way comparison of masking strategies (random, 2D block, context-block, text-side span, and a per-sample mixture) under identical compute and architecture, evaluated under one unified protocol. We treat the mask as the independent variable.

III. METHOD

Setup: We train nano4M, a four-modality FourM model, on the course-provided CLEVR dataset (50k train, 5k val, 5k test scenes). The four modalities are tokenised RGB, depth, and surface-normal maps (Cosmos image tokenizer [6], $16 \times 16 = 256$ tokens per modality) and a scene caption (scene_desc) tokenised with GPT-2 to a maximum of 256 tokens. The model is a FourM encoder-decoder transformer with encoder depth 6, decoder depth 6 and model dimension 512. Every variant is trained for 5 billion tokens at batch size 256 (approximately 38 000 iterations), with the same optimizer settings, the same data, and matched random seeds. Only the masking strategy varies. Training loss curves are shown in Appendix D.

Token-budget allocation: For each training sample, two integers are drawn independently and uniformly from $[1, 128]$: one is the total encoder-input budget, the other is the total decoder-target budget. A per-sample Dirichlet then splits each budget across the four modalities. Within each modality, the masking strategy decides which specific positions become encoder input and which become decoder target. Positions not selected by either are unused for that sample. The two integer ranges, batch size, and total token budget are identical across all variants and ablations.

The five masking strategies: Figure 1 illustrates V0, V1, V2, V4 on the image-token grid; Figure 2 illustrates V3 on the text-token sequence.

V0 baseline. Image-side input and target positions are sampled uniformly at random from the 256-position grid; text-side positions are also random within scene_desc. This is the original 4M [1] masking recipe and the reference all other variants are compared against.

V1 block. For each image modality, random $s \times s$ blocks (with s drawn from $\{2, 3, 4\}$ per sample) are placed on the 16×16 grid until enough positions are covered for both budgets; the input and target positions for that modality are then drawn from inside the block region. Removes the local-neighbour shortcut by clustering both input and target into block-shaped regions. Text-side masking stays at V0.

V2 ctxblock. For each image modality, a single $s \times s$ visible block (s drawn from $\{4, 5, 6\}$ per sample, block position sampled independently per modality) is kept as encoder input; target tokens are then scattered outside the block. With around 10% of every image-side modality visible on average, the caption becomes the only substantially-visible scaffold and the model is forced to rely on cross-modal context. Text-side masking stays at V0.

V3 span. The text modality gets contiguous-span masking. Span lengths are drawn from a geometric distribution with mean $\mu = 3$ and placed at random non-overlapping positions in scene_desc until the target budget is filled. The mean span length was calibrated empirically against CLEVR: the median GPT-2 tokenised length of a “Field: value” phrase in scene_desc is exactly 3 tokens, so a typical span hides one whole semantic field such as “Color: yellow” or “Shape: cube”. Image modalities keep V0 random masking.

V4 mixed. Per training sample, a Bernoulli(0.5) coin is flipped: heads applies V0 masking to that sample, tails applies V2. The model sees both objectives during training. The mixture ratio was set at the TA’s suggestion.

Ablations: We further train six controlled ablations to test the sensitivity of V1 and V2 to their block-size hyperparameter: V1 with the block size fixed at 2, 3, and 4 individually (block_b2, block_b3, block_b4), and V2 with the ctxblock size fixed at 4, 5, and 6 individually (ctxblock_c4, ctxblock_c5, ctxblock_c6). Each ablation is a full retraining run from scratch with the corresponding size held fixed instead of sampled, isolating the effect of the size-sampling itself.

IV. EXPERIMENTS

Setup: We evaluate every variant under one unified protocol on a held-out subset of the CLEVR val split: the first 500 sorted-by-filename samples. Each variant is evaluated three times with seeds 0, 1, 2 (same checkpoint, same 500 samples, only the stochastic token-sampling RNG differs), and the per-sample metric values are averaged across seeds. The evaluation covers two task families. *Phase A reconstruction:* hide one modality and rebuild it from the other three. *Phase B cross-modal generation:* give the model a single modality and generate another from scratch, in both directions (image to caption, caption to image). All generation uses an 8-step parallel iterative decoder [4], matched across variants.

Metrics: We report seven metrics covering complementary aspects of model behaviour. *Parser field accuracy* (sd/pos): the generated caption is parsed with a regex and the per-object position fields are compared exactly to ground truth. *CIDEr-D* [7]: standard TF-IDF weighted n-gram overlap, computed on Phase A scene_desc reconstruction (*CIDEr A*) and on Phase B.1 image-to-caption (*CIDEr B*). *Depth AbsRel* and *RGB SSIM* [8]: pixel-level reconstruction quality on Phase A. *LLM judge:* Qwen 2.5-7B-Instruct [9] scores semantic alignment between generated and ground-truth captions on Phase B.1; meaning-aware, less sensitive to wording than CIDEr. *Object-detection F1:* GroundingDINO-tiny [10] runs on the generated image with the objects parsed from the ground-truth caption as queries; F1 is computed over the resulting set. Phase B.2 caption-to-image only, since CIDEr and the LLM judge already cover the captioning side.

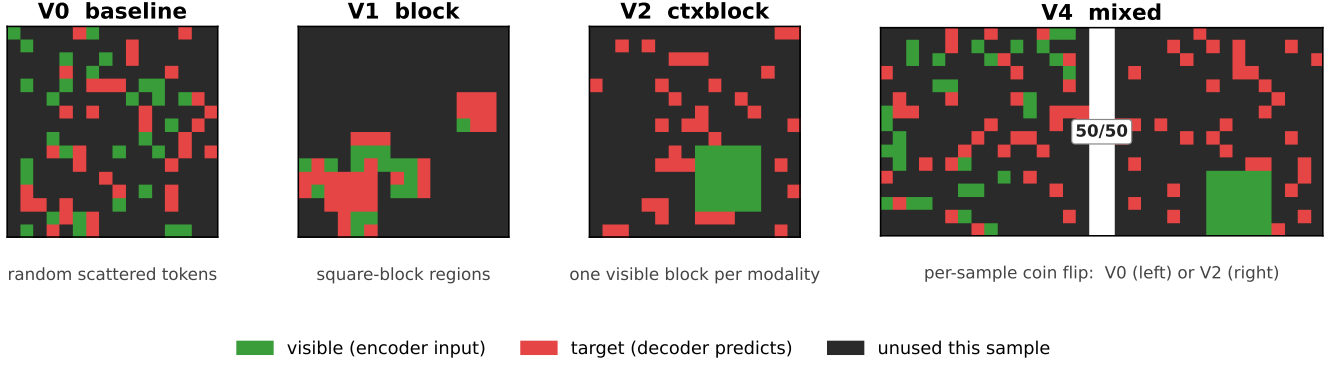


Figure 1. Image-side masking strategies. Green = encoder input (visible to the model), red = decoder target (the model must predict it), dark grey = unused for this sample. V0 scatters input and target positions uniformly. V1 clusters them inside square block regions. V2 keeps one $s \times s$ visible block per modality and scatters targets outside. V4 alternates V0 and V2 per sample (50/50). See Figure 2 for V3, which masks the text modality.

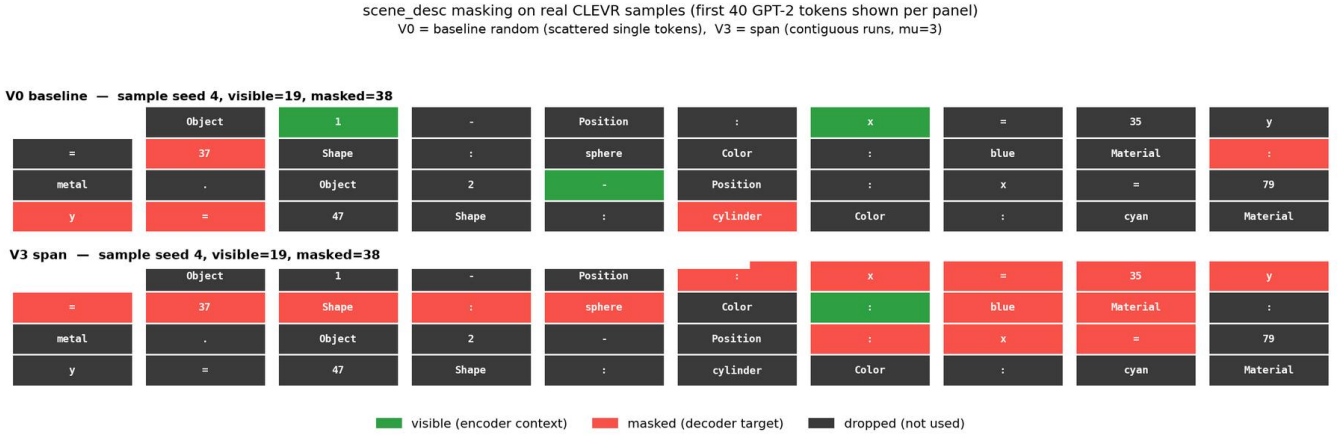


Figure 2. V3 span masking on real CLEVR scene_desc tokens (first 40 GPT-2 tokens shown, sample seed 4). Top: V0 baseline scatters single-token targets at random positions. Bottom: V3 span produces contiguous runs of red target tokens, calibrated so a typical run covers about one “Field: value” phrase.

Headline results: Table I reports the 3-seed mean of every metric for every main variant. Four findings stand out.

Finding 1: V2 ctxblock wins on meaning-aware and object-grounded metrics in both directions: V2 takes the highest parser field accuracy by a large margin (0.519 against 0.216 baseline, a $2.4\times$ relative improvement at placing objects), the highest CIDEr A on Phase A reconstruction (1.54 against 1.02), the highest LLM judge on Phase B.1 (0.122 against 0.103) and the highest object-detection F1 on Phase B.2 (0.659 against 0.633). The three metrics that actually look at semantic or object-level content (the parser, the LLM judge and the detector) all agree V2 is the strongest variant. The mechanism is consistent with the training objective: with around 90% of every image-side modality hidden during training, the only substantially-visible scaffold is the caption, so the model is pushed to bind image tokens to text, and that binding pays off at generation time.

Finding 2: V1 block wins pixel-level reconstruction: V1 achieves the lowest depth AbsRel (0.027 against 0.029) and the highest RGB SSIM (0.888 against 0.883). Block masking removes the local-neighbour shortcut by forcing the model to predict from non-adjacent context, which is what sharper depth and RGB textures need.

Finding 3: V4 mixed wins lexical caption similarity in image-to-caption: V4 wins CIDEr B with 0.94, beating both of its parents (V0 = 0.89, V2 = 0.78), and beating V2 in particular, which is the worst on this lexical metric. Image-to-caption needs both broad token coverage (V0’s strength) and cross-modal binding (V2’s strength); neither pure mask wins this row alone, the hybrid does. V4 makes the metric-specific framing explicit: V2 takes the semantic and object-grounded angle, V4 takes the lexical angle.

Finding 4: V3 span ties with V0 baseline: On every text-side metric, V3 sits within one seed standard deviation of V0: parse rate 0.998 vs 0.999, CIDEr A 1.00 vs 1.02,

Table I

HEADLINE METRICS ACROSS THE FIVE MAIN VARIANTS. MEAN OVER 3 SEEDS, 500 CLEVR VAL SAMPLES EACH. $\uparrow$ HIGHER IS BETTER, $\downarrow$ LOWER IS BETTER. BOLD = COLUMN-BEST WITHIN THE MAIN FIVE VARIANTS. PER-SEED STANDARD DEVIATIONS ARE ≤ 0.012 FOR SD/POS, DEPTH, RGB SSIM, LLM AND DINO F1, AND UP TO 0.13 FOR CIDER A/B (TF-IDF WEIGHTING IS SENSITIVE TO PER-SAMPLE N-GRAM DISTRIBUTIONS); THE HEADLINE GAPS WE REPORT ARE WELL OUTSIDE THIS SEED VARIABILITY.

Variant	Reconstruction (Phase A)				Image $\rightarrow$ caption (B.1)		Caption $\rightarrow$ image (B.2)
	sd/pos $\uparrow$	CIDEr A $\uparrow$	depth $\downarrow$	RGB SSIM $\uparrow$	CIDEr B $\uparrow$	LLM $\uparrow$	DINO F1 $\uparrow$
V0 baseline	0.216	1.02	0.029	0.883	0.89	0.103	0.633
V1 block	0.211	0.99	0.027	0.888	0.90	0.106	0.649
V2 ctxblock	0.519	1.54	0.029	0.886	0.78	0.122	0.659
V3 span	0.220	1.00	0.029	0.884	0.82	0.097	0.643
V4 mixed	0.243	1.10	0.029	0.883	0.94	0.109	0.653

sd/pos 0.220 vs 0.216, LLM judge 0.097 vs 0.103, CIDEr B 0.82 vs 0.89. The only metric where V3 nudges ahead beyond seed noise is parser/color (0.058 vs 0.048). We report this as a negative result: text-side span masking, even when calibrated to the data ($\mu = 3$ matches the median CLEVR Field:value tokenised length), did not meaningfully improve text generation over V0 at our 5-billion-token budget. Our reading is threefold. First, CLEVR’s per-field vocabulary is small (three shapes, eight colors, two materials) and V0 already saturates these marginals. Second, text is one of four modalities in the loss, so the masking-strategy gradient is diluted by the three image modalities. Third, the bottleneck is visual grounding (unchanged by V3) rather than text-side prediction. None of this invalidates the V3 design; it locates the bottleneck.

Ablations: The six ablations confirm that V1’s pixel-reconstruction win is robust to block size (depth AbsRel = 0.027 at block_b3 and block_b4, 0.028 at block_b2) but V2’s wins depend on the size-sampling: every ctxblock_c4 / c5 / c6 ablation underperforms V2 main on field accuracy, CIDEr A, LLM judge and DINO F1. The random sampling across $\{4, 5, 6\}$, not any single fixed size, is what makes V2 a robust cross-modal binder.

Qualitative results: Figure 3 shows Phase A RGB reconstruction on sample 319 (a 7-object scene). All variants receive the same depth, normals and caption inputs. V2’s reconstruction has the layout closest to the ground truth, consistent with its dominance on the semantic metrics.

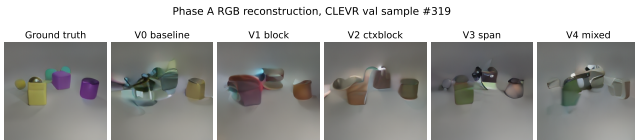


Figure 3. Phase A RGB reconstruction on CLEVR val sample 319. All variants receive the same depth, normals and caption inputs. V2’s layout is visually closest to the ground truth, matching its dominance on the semantic metrics in Table I.

V. CONCLUSION AND LIMITATIONS

We studied how the masking strategy at pre-training shapes what a small four-modality model learns. Under matched architecture, data, compute and seeds, three different masks produced three different specialists: V2 context-block wins the meaning-aware and object-grounded metrics in both generation directions, V1 block wins pixel-level reconstruction, and V4 mixed (V0 + V2) wins lexical caption similarity in image-to-caption. V3 text-side span masking ties with the random baseline, an honest negative result we attribute to visual grounding being the bottleneck in this setup. The mask, in our setting, is the objective.

Limitations: Our experiments are confined to CLEVR, which has a small per-field vocabulary; rankings may shift on richer language data. Three seeds per variant suffices to confirm the large gaps we report (V2’s $2.4\times$ field-accuracy lead) are outside seed variability, but is not enough to claim significance for the smaller gaps. The per-sample LLM-judge / parser agreement on Phase B.1 is low (around 0.10); we therefore treat the LLM judge as a supplementary signal, not the primary one for text generation.

Future work: Two natural directions. First, an alternative V3 implementation using canonical T5-style span corruption [5] under a fine-tuning regime, since T5 was designed as a pre-train-then-fine-tune objective. Second, exploring alternative iterative-decoding orders (centre-outward spiral, anchor-then-fill, raster) instead of the current confidence-first scheme. The headline table is reproducible via the provided `run_evaluation.py`.

VI. INDIVIDUAL CONTRIBUTIONS

Per the Team 11 Extension Plan: **Gabriel Chidiac** (V0, V2 and its ablations), **Nacer Saadani-Hassani** (V1 and its ablations), **Ricardo Habib** (V3, V4, evaluation extensions), **Ralph Chidiac** (evaluation pipeline and extensions). All authors contributed to the report.

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APPENDIX A.

FULL RESULTS INCLUDING THE SIX ABLATIONS

Table II extends Table I with the six controlled ablations introduced in Section III. The ablations train V1 with the block size fixed at 2, 3, and 4 individually, and V2 with the ctxblock size fixed at 4, 5, and 6 individually. Each ablation is a full retraining run from scratch with the corresponding size held fixed instead of sampled.

Two observations from the ablation block. First, V1’s pixel-reconstruction win is broadly robust to the block size: depth AbsRel is 0.027 at block_b3 and block_b4, 0.028 at block_b2; RGB SSIM is also stable across block_b2, block_b3 and block_b4 (with block_b4 even slightly above V1 main at 0.892 SSIM). Second, V2’s wins are fragile to fixing the ctxblock size: every ctxblock_c4 / c5 / c6 ablation underperforms V2 main on field accuracy, CIDEr A, LLM judge and DINO F1, often substantially (e.g. CIDEr B drops from V2 main’s 0.78 to 0.32 at ctxblock_c5 and 0.30 at ctxblock_c6). The random sampling across {4, 5, 6}, not any single fixed size, is what makes V2 a robust cross-modal binder.

APPENDIX B.

PER-MODALITY QUALITATIVE RECONSTRUCTION

Figure 3 in the main report shows Phase A RGB reconstruction for sample 319 across the five main variants. Figures 4 and 5 extend this to depth and surface-normal reconstruction on the same sample with the same inputs.

APPENDIX C.

PHASE A CAPTION RECONSTRUCTION ON SAMPLE 319

Companion to Figures 3, 4 and 5: the text caption produced by each main variant for the same CLEVR val sample 319, given RGB, depth and surface-normal inputs. Captions are reproduced verbatim from the cached eval outputs.

Ground truth. Object 1 - Position: x=33 y=50 Shape: cube Color: yellow Material: rubber. Object 2 - Position: x=58 y=30 Shape: sphere Color: cyan Material: rubber. Object 3 - Position: x=55 y=37 Shape: cube Color: purple Material: rubber. Object 4 - Position: x=82 y=46 Shape: cylinder Color: purple Material: metal. Object 5 - Position: x=30 y=42 Shape: sphere Color: yellow Material: metal. Object 6 - Position: x=45 y=32 Shape: sphere Color: cyan Material: rubber. Object 7 - Position: x=16 y=41 Shape: sphere Color: yellow Material: rubber.

V0 baseline. Object 1 - Position: x=16 y=32 Shape: cylinder Color: purple Material: metal. Object 2 - Position: x=34 y=40 Shape: sphere Color: cyan Material: metal. Object 3 - Position: x=82 y=44 Shape: sphere Color: green Material: metal. Object 4 - Position: x=83 y=50 Shape: cylinder Color: blue Material: rubber. Object 5 - Position: x=47 y=38 Shape: sphere Color: brown Material: rubber. Object 6 - Position: x=18 y=52 Shape: cube Color: brown Material: rubber. Object 7 - Position: x=49 y=49 Shape: sphere Color: purple Material: metal.

V1 block. Object 1 - Position: x=83 y=39 Shape: cube Color: green Material: metal. Object 2 - Position: x=30 y=47 Shape: sphere Color: yellow Material: metal. Object 3 - Position: x=30 y=34 Shape: cylinder Color: yellow Material: metal. Object 4 - Position:

x=29 y=32 Shape: sphere Color: brown Material: rubber. Object 5 - Position: x=80 y=44 Shape: cylinder Color: gray Material: rubber. Object 6 - Position: x=79 y=32 Shape: cube Color: brown Material: metal. Object 7 - Position: x=69 y=41 Shape: cylinder Color: brown Material: rubber. Object 8 - : x=30 y=

V2 ctxblock. Object 1 - Position: x=82 y=50 Shape: sphere Color: green Material: rubber. Object 2 - Position: x=34 y=43 Shape: cylinder Color: green Material: metal. Object 3 - Position: x=44 y=31 Shape: sphere Color: yellow Material: rubber. Object 4 - Position: x=55 y=36 Shape: cube Color: yellow Material: rubber. Object 5 - Position: x=15 y=42 Shape: sphere Color: gray Material: rubber. Object 6 - Position: x=45 y=31 Shape: sphere Color: cyan Material: rubber. Object 7 - Position: x=44 y=31 Shape: sphere Color: blue Material: rubber.

V3 span. Object 1 - Position: x=38 y=30 Shape: cube Color: brown Material: rubber. Object 2 - Position: x=58 y=39 Shape: cylinder Color: yellow Material: rubber. Object 3 - Position: x=83 y=33 Shape: cube Color: green Material: metal. Object 4 - Position: x=69 y=29 Shape: sphere Color: purple Material: rubber. Object 5 - Position: x=30 y=35 Shape: sphere Color: green Material: rubber. Object 6 - Position: x=44 y=29 Shape: sphere Color: purple Material: rubber. -=16 8

V4 mixed. Object 1 - Position: x=56 y=51 Shape: cylinder Color: cyan Material: rubber. Object 2 - Position: x=47 y=40 Shape: sphere Color: brown Material: rubber. Object 3 - Position: x=29 y=37 Shape: cylinder Color: blue Material: rubber. Object 4 - Position: x=14 y=42 Shape: cube Color: blue Material: metal. Object 5 - Position: x=59 y=33 Shape: sphere Color: cyan Material: rubber. Object 6 - Position: x=82 y=35 Shape: cylinder Color: gray Material: rubber.

Observations consistent with the headline metrics in Table I: V0 and V2 produce clean seven-object captions matching the ground-truth object count; V1 emits seven objects followed by a fragmented eighth; V4 stops cleanly at six objects (one short of the ground truth); V3 reaches six objects then emits a garbled tail. Position values disagree with ground truth across all variants, consistent with the low absolute sd/pos numbers in Table I, but the structural template (Object N - Position: $x = \dots y = \dots$ Shape: $\dots$ Color: $\dots$ Material: $\dots$) is recovered cleanly by every variant. V4’s short-by-one output is task-specific to this Phase A reconstruction (caption rebuilt from RGB + depth + normals) and does not contradict V4’s win on the separate Phase B.1 lexical metric (CIDEr B in Table I), which scores image-to-caption generation from scratch.

Table II

FULL RESULTS ACROSS THE FIVE MAIN VARIANTS AND THE SIX BLOCK-SIZE ABLATIONS. MEAN OVER 3 SEEDS, 500 CLEVR VAL SAMPLES EACH. SAME COLUMN ORDER AS TABLE I. THE MIDRULE SEPARATES THE MAIN VARIANTS (TOP) FROM THE ABLATIONS (BOTTOM).

Variant	Reconstruction (Phase A)				Image $\rightarrow$ caption (B.1)		Caption $\rightarrow$ image (B.2)
	sd/pos $\uparrow$	CIDEr A $\uparrow$	depth $\downarrow$	RGB SSIM $\uparrow$	CIDEr B $\uparrow$	LLM $\uparrow$	DINO F1 $\uparrow$
V0 baseline	0.216	1.02	0.029	0.883	0.89	0.103	0.633
V1 block	0.211	0.99	0.027	0.888	0.90	0.106	0.649
V2 ctxblock	0.519	1.54	0.029	0.886	0.78	0.122	0.659
V3 span	0.220	1.00	0.029	0.884	0.82	0.097	0.643
V4 mixed	0.243	1.10	0.029	0.883	0.94	0.109	0.653
block_b2	0.217	1.03	0.028	0.888	0.90	0.104	0.643
block_b3	0.215	1.06	0.027	0.889	0.87	0.109	0.655
block_b4	0.216	1.00	0.027	0.892	0.82	0.113	0.652
ctxblock_c4	0.334	0.69	0.032	0.882	0.58	0.079	0.644
ctxblock_c5	0.447	0.78	0.031	0.883	0.32	0.113	0.647
ctxblock_c6	0.208	0.86	0.030	0.885	0.30	0.088	0.582

Phase A depth reconstruction, CLEVR val sample #319

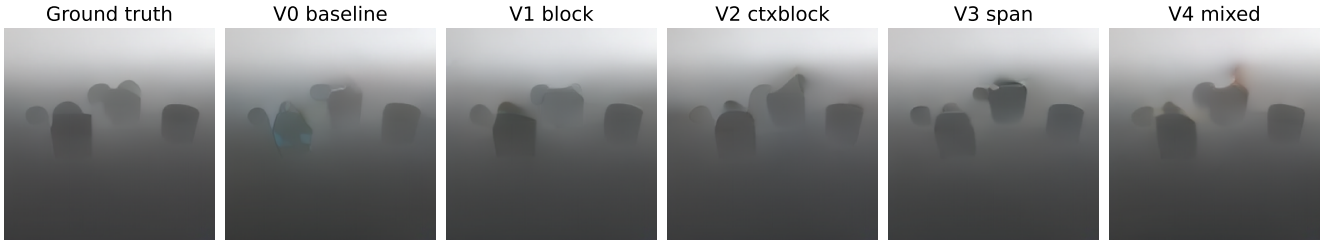


Figure 4. Phase A depth reconstruction on CLEVR val sample 319. All five variants receive the same RGB, normals and caption inputs. The depth maps recover the scene’s surface structure across variants; V1’s reconstruction is slightly sharper at object boundaries, consistent with its column-best depth AbsRel in Table I.

Phase A surface-normal reconstruction, CLEVR val sample #319

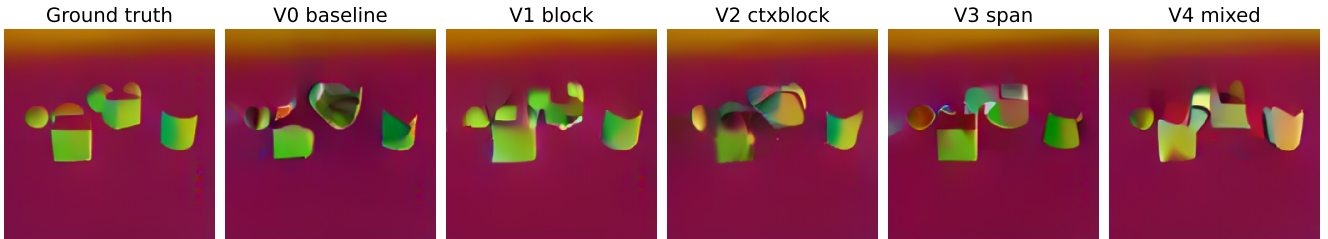


Figure 5. Phase A surface-normal reconstruction on CLEVR val sample 319. All five variants receive the same RGB, depth and caption inputs. Normal maps encode the surface orientation per pixel as RGB; correctly oriented surfaces match the ground-truth colour pattern.

APPENDIX D. TRAINING DYNAMICS

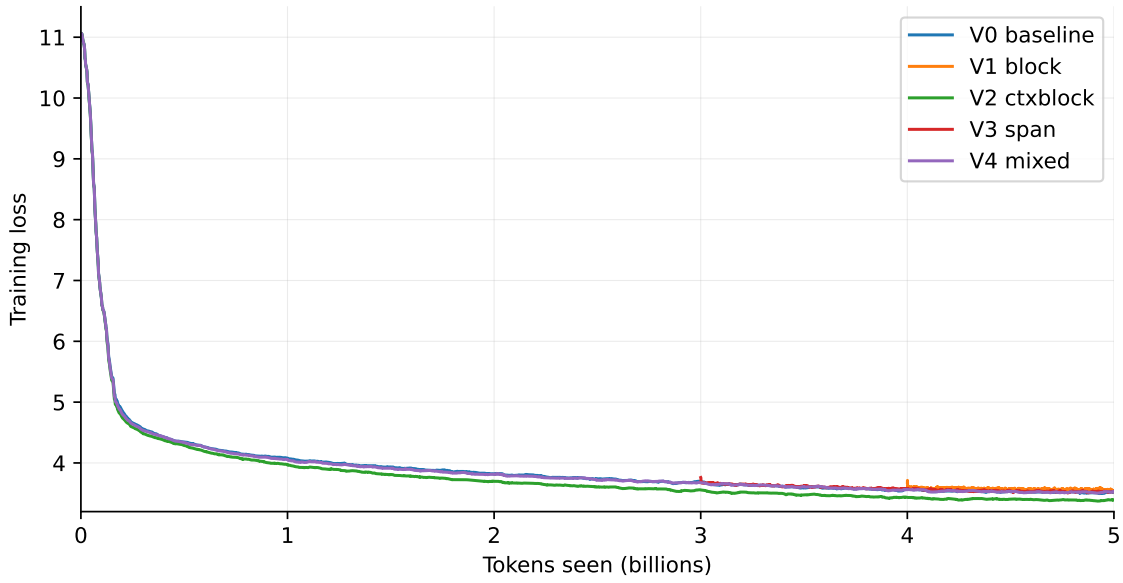


Figure 6. Training loss for the five main variants over the full 5-billion-token budget (38 146 steps, batch size 256). All five curves track each other closely, confirming the matched-conditions design. V2 ctxblock converges to a slightly lower training loss than the other four. Curves are smoothed with a 50-step rolling mean for readability.

The headline evaluations in Section IV rest on the claim that the five main variants were trained under matched conditions, with only the masking strategy varying between them. Figure 6 confirms this visually: training loss for all five main variants follows the same shape across the full 5-billion-token budget. No variant diverged, oscillated, or stalled. V2 ctxblock converges to a marginally lower value than the other four, which we attribute to its training objective being slightly easier to fit (each sample has a single small visible image-block plus many target tokens, so the model has both a clear scaffold to condition on and a large prediction signal at every step). This is not the same as downstream quality, which Table I measures separately.